Haysun Hannah Choi

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Education

Yale University Sep 2021 – current
PhD Candidate in Ecology & Evolutionary Biology (E&EB)
Advisor: Richard O. Prum

University of Utah Sep 2013 – May 2018
Honors BS in Applied Mathematics, minor in Chemistry; GPA: 3.8/4.0

Scholarships and Grants

Dissertation Improvement Grant Yale Institute for Biospheric Studies ($5000) 2024
Public Humanities Grant Yale Environmental Humanities ($2500) 2023 – 2024
Collections Engagement Grant The Utah Museum of Fine Arts ($4400) 2020
NSF Research Experience for Undergraduates University of Utah Mathematics ($1500) Fall 2017
NSF Research Experience for Undergraduates University of Utah Mathematics ($1500) Spring 2018
Outdoor Education Continuing Learning Grant University of Utah Outdoor Program ($1000) Spring 2017
Honors at Entrance University of Utah Admissions (Tuition) 2013 – 2017
ACCESS Women in Science University of Utah College of Science ($5000) Summer 2013
Regents' Scholarship Utah System of Higher Education ($4000) 2013 –2014

Publications

Choi, H.H. and Prum, R.O. Sexual Selection in the Price Equation. *Journal of Evolutionary Biology.* [In review]
Choi, H.H. and Prum, R.O. A Darwinian, Quantitative Genetics Framework for Alternative Mechanisms of Intersexual Selection and Sexual Conflict over Mating. [In prep.]

Presentations and Posters*

Hyperuniformity in Non-Iridescent, Avian Photonic Glasses* June 2026
Choi, H.H., Prum, R.O., and Saranathan V. Bioinspired Materials GRC, Les Diablerets, CH
A Darwinian Framework for Intersexual Mechanisms of Evolution over Mating Feb 2026
Choi, H.H. and Prum, R.O. Edwards Lab, Harvard University, Boston, US
Modeling the Development of Structural Color: Applications of Cahn-Hillard and Persistent Homology May 2025
Choi, H.H., Prum, R.O., Ma, T., and Pedersen M. C. Active Matter Lab, Niels Bohr Institute, Copenhagen, DK
Reproduction in the Price Equation Dec 2024
Choi, H.H. and Prum, R.O. E&EB Graduate Research Symposium, Yale, New Haven, US
A Unified Framework for Mechanisms of Mate Choice, Sexual Conflict, and Intrasexual Competition* July 2024
Choi, H.H. and Prum, R.O. Society for the Study of Evolution Meeting, Montreal, CA
Sexual Selection in the Price Equation* May 2024
Choi, H.H. and Prum, R.O. Sexual Selection in a Changing World Workshop, Erice, IT
Eclectic Forms Dec 2021
Choi, H.H. Environmental Humanities Symposium, Yale, New Haven, US
Form-ing Curiosity Feb 2020
Choi, H.H., Liechty, S. S., Rose, C., and Mershin, A. Center for Bits & Atoms, MIT, Boston, US
A Mathematical Approach to Human Experience May 2018
Choi, H.H., Lawley, S., and Seger, J. Honors College Symposium, University of Utah, Salt Lake City, US
Optimal Predator Search Strategies Using Olfaction and Mean First Passage Time May 2018
Choi, H.H. and Lawley, S. Senior Research Symposium, University of Utah, Salt Lake City, US

External Academic Training

Visiting Researcher Biophysics & Biocomplexity Group, Niels Bohr Institute — Spring 2025
Participant Sexual Selection in a Changing World Workshop, Università degli Studi di Padova — Summer 2024
Participant Modeling & Theory in Population Biology Workshop, Banff International Research Station — Spring 2024
Student Evolutionary Quantitative Genetics Workshop, University of Washington — Summer 2022
Coordinator Multivariate Evolutionary Theory Reading Group, University of Utah — Summer 2022
Student Network Science & Complexity Reading Group, Northeastern University — Spring 2022

Community Leadership and Outreach

Graduate Mentor Yale E&EB — 2023 – 2024
Hutchinson Speaker Coordinator Yale E&EB — Fall 2024
Elected Graduate Student Representative Yale E&EB — 2022 – 2023
DEI Committee Member Yale E&EB — 2022 – 2023
Conference Delegate Graduate Biology Meeting for Columbia, Rutgers, Princeton, UPenn, and Yale — Spring 2023
Graduate Professional Fellow in Light Measurement Yale Center for British Art — Spring 2023
Reader of Fiction The Yale Review — 2021 – 2022
Assistant Editor University of Utah Special Collections — 2021
Equipment Studio Monitor University of Utah Book Arts Program — 2017 – 2019
Grader; Tutor University of Utah Mathematics — 2018 – 2019
Backcountry Trip Leader; Trainer University of Utah Outdoor Program — 2014 – 2018
D.C. National Scholar University of Utah Hinckley Institute of Politics — Summer 2016

Collaborations and Creative Projects

Workshop Consequences of Environmental Stochasticity for Spatial Dynamics — Spring 2022
National Science Foundation, Yale Fine Arts, Yale E&EB, Princeton E&EB
Journal; Short Film Abandon Shape; From the Marrow — Fall 2021
University of Utah Honors College, University of Utah Special Collections, Utah Museum of Fine Arts
Artists Book Translational Reconciliation — Summer 2021
University of Utah Book Arts Program
Workshop Honing Curiosity — Winter 2020
MIT Center for Bits & Atoms
Undergraduate Thesis A Mathematical Approach to Human Experience — May 2018
University of Utah Honors College

Teaching Fellowships & Assistantships

Ornithology Lecture and Lab, Yale E&EB — Spring 2023
Introduction to Evolution and Ecology, Yale E&EB — Spring 2022, Fall 2024
Terrestrial Arthropods Lecture and Lab, Yale E&EB — Fall 2021
Biology of Variation, University of Utah Biology — Fall 2019
Rock Climbing I-II, University of Utah Parks, Recreation, & Tourism — Fall 2016

Professional Experience

Sous Chef Nohm, Salt Lake City, US — 2019 – 2021
Logistics Coordinator; Assistant Instructor Northwest Outward Bound School, Mazama, US — 2018 – 2019
Medical Clerk; First Responder Alta Ski Resort, Sandy, US — 2016 – 2018